

Connection To azure DB

Default Connection String db:

```
"Server=tcp:ajial-db-server.database.windows.net,1433;Initial Catalog=Ajial-db-dev;Persist  
Security Info=False;User  
ID=ajial-db-server;Password=4179@12345@AAST@;MultipleActiveResultSets=False;Encrypt=  
True;TrustServerCertificate=False;Connection Timeout=30;"
```

```
onnection(url, "ajial-db-server@ajial-db-server", {your_password});
```

Register Parent

Test :Swagger:

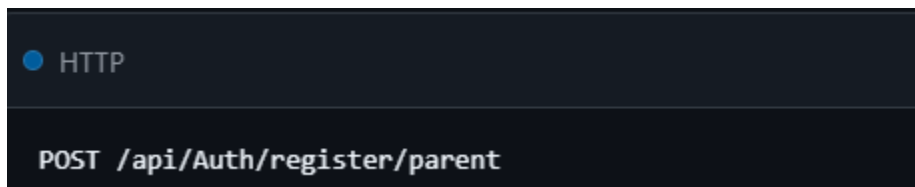
<https://ajial-api-dev-dvg9hfgtdgewekcv.westeurope-01.azurewebsites.net/swagger/index.html>

Test :Postman:

<https://ajial-api-dev-dvg9hfgtdgewekcv.westeurope-01.azurewebsites.net/api/Auth/register/parent>

1. Register Parent

Creates a new parent user account in the system.



Request Headers

Content-Type: application/json

Accept: application/json

Request Body

```
{
  "fullName": "string",    // Required, 3-100 characters
  "username": "string",    // Required, 3-50 characters, unique, alphanumeric + underscore
  "email": "string",       // Required, valid email format, unique
  "password": "string",    // Required, min 8 chars, must contain uppercase, lowercase, number,
                           // special char
  "cityId": 0,             // Required, integer, must exist in Cities table (1-5)
  "dateOfBirth": "YYYY-MM-DD", // Required, user must be 18+ years old
  "gender": 0              // Required, 1 = Male/Father, 2 = Female/Mother
}
```

Field Validation Rules			
Field	Type	Required	Validation Rules
fullName	string	✓ Yes	• 3-100 characters • No leading/trailing spaces • Can contain letters, spaces, Arabic characters
username	string	✓ Yes	• 3-50 characters • Must be unique • Alphanumeric + underscore only • No spaces
email	string	✓ Yes	• Valid email format • Must be unique • Max 100 characters
password	string	✓ Yes	• Minimum 8 characters • Must contain: - At least 1 uppercase letter - At least 1 lowercase letter - At least 1 number - At least 1 special character (@\$!%*?&#)

Examples

Example Value | Schema

```
{
  "fullName": "Ahmed Mohamed",
  "username": "ahmed2025",
  "email": "ahmed@example.com",
  "password": "Ahmed@123",
  "cityId": 1,
  "dateOfBirth": "1985-03-20",
  "gender": 1
}
```

Responses

Request URL

<https://ajial-api-dev-dvg9hfgtdgewekcv.westeurope-01.azurewebsites.net/api/Auth/register/parent>

Code:201 :Created

Response body

```
{
  "success": true,
  "message": "تم إنشاء الحساب بنجاح",
  "data": {
    "userId": "8dc76a4b-f02d-4b04-8283-592e8aad5a11",
    "parentId": "cd3079e0-6ffb-42b9-82b4-b7b23a7e8d66",
    "fullName": "Ahmed Mohamed",
    "username": "ahmed2025",
    "email": "ahmed@example.com",
    "message": "تم إنشاء الحساب بنجاح",
    "createdAt": "2025-10-26T13:18:31.0742463Z"
  },
  "errors": []
}
```

Response headers

```
access-control-allow-origin: *
content-type: application/json; charset=utf-8
date: Sun, 26 Oct 2025 13:18:31 GMT
location: https://ajial-api-dev-dvg9hfgtdgewekcv.westeurope-01.azurewebsites.net/api/Auth/register/parent
server: Microsoft-IIS/10.0
transfer-encoding: chunked
x-powered-by: ASP.NET
```

Error Response (400 Bad Request)

```
{
  "success": false,
  "message": "فشل في إنشاء الحساب",
  "data": null,
  "errors": [
    "اسم المستخدم موجود بالفعل",
    "البريد الإلكتروني مستخدم بالفعل"
  ]
}
```

Code: 500 :Internal Server Error

Dart Data Moduls

- Parent Registration Request

```
class ParentRegisterRequest {
    final String fullName;
    final String username;
    final String email;
    final String password;
    final int cityId;
    final String dateOfBirth; // Format: "YYYY-MM-DD"
    final int gender;        // 1 = Male, 2 = Female

    ParentRegisterRequest({
        required this.fullName,
        required this.username,
        required this.email,
        required this.password,
        required this.cityId,
        required this.dateOfBirth,
        required this.gender,
    });

    Map<String, dynamic> toJson() => {
        'fullName': fullName,
        'username': username,
        'email': email,
        'password': password,
        'cityId': cityId,
        'dateOfBirth': dateOfBirth,
        'gender': gender,
    };
}
```


- Parent Registration Response

```
class ParentRegisterResponse {
  final bool success;
  final String message;
  final ParentRegisterData? data;
  final List<String> errors;

  ParentRegisterResponse({
    required this.success,
    required this.message,
    this.data,
    required this.errors,
  });

  factory ParentRegisterResponse.fromJson(Map<String, dynamic> json) {
    return ParentRegisterResponse(
      success: json['success'] as bool,
      message: json['message'] as String,
      data: json['data'] != null
        ? ParentRegisterData.fromJson(json['data'])
        : null,
      errors: (json['errors'] as List<dynamic>?)
        ?.map((e) => e.toString())
        .toList() ?? [],
    );
  }
}

class ParentRegisterData {
  final String userId;
  final String parentId;
  final String fullName;
  final String username;
  final String email;
  final String message;
  final DateTime createdAt;

  ParentRegisterData({
    required this.userId,
    required this.parentId,
    required this.fullName,
    required this.username,
    required this.email,
    required this.message,
    required this.createdAt,
  });

  factory ParentRegisterData.fromJson(Map<String, dynamic> json) {
    return ParentRegisterData(
      userId: json['userId'] as String,
      parentId: json['parentId'] as String,
      fullName: json['fullName'] as String,
      username: json['username'] as String,
      email: json['email'] as String,
      message: json['message'] as String,
      createdAt: DateTime.parse(json['createdAt'] as String),
    );
  }
}
```

- Gender Enum

```
enum Gender {  
  male(1),  
  female(2);  
  
  final int value;  
  const Gender(this.value);  
  
  static Gender fromValue(int value) {  
    return Gender.values.firstWhere(  
      (e) => e.value == value,  
      orElse: () => throw ArgumentError('Invalid gender value: $value'),  
    );  
  }  
}
```

- **City Model**

```
class City {  
    final int id;  
    final String name;  
    final String nameAr;  
    final bool isActive;  
  
    City(  
        required this.id,  
        required this.name,  
        required this.nameAr,  
        required this.isActive,  
    );  
  
    factory City.fromJson(Map<String, dynamic> json) {  
        return City(  
            id: json['id'] as int,  
            name: json['name'] as String,  
            nameAr: json['nameAr'] as String,  
            isActive: json['isActive'] as bool,  
        );  
    }  
}
```

Available Cities for now:

ID	Name (English)	Name (Arabic)
1	Cairo	القاهرة
2	Alexandria	الإسكندرية
3	Giza	الجيزة
4	Shubra El Kheima	شبرا الخيمة
5	Port Said	بورسعيد

Test Data Sets

Valid Test Cases

// Test Case 1: Father from Cairo

```
{  
  "fullName": "Ahmed Mohamed",  
  "username": "ahmed_test_001",  
  "email": "ahmed001@test.com",  
  "password": "Ahmed@123",  
  "cityId": 1,  
  "dateOfBirth": "1985-01-15",  
  "gender": 1  
}
```

// Test Case 2: Mother from Alexandria

```
{  
  "fullName": "Fatima Hassan",  
  "username": "fatima_test_002",  
  "email": "fatima002@test.com",  
  "password": "Fatima@456",  
  "cityId": 2,  
  "dateOfBirth": "1990-06-20",  
  "gender": 2  
}
```

// Test Case 3: Arabic Name

```
{  
  "fullName": "محمد أحمد علي",  
  "username": "mohamed_test_003",  
  "email": "mohamed003@test.com",  
  "password": "Mohamed@789",  
  "cityId": 3,  
  "dateOfBirth": "1988-12-10",  
  "gender": 1  
}
```

Invalid Test Cases (Should Fail)

// Test Case 1: Under 18

```
{
  "fullName": "Young User",
  "username": "young_user",
  "email": "young@test.com",
  "password": "Young@123",
  "cityId": 1,
  "dateOfBirth": "2010-01-01", // ✗ Too young
  "gender": 1
}
```

// Test Case 2: Weak Password

```
{
  "fullName": "Test User",
  "username": "test_weak_pass",
  "email": "weakpass@test.com",
  "password": "123456", // ✗ No uppercase, special char
  "cityId": 1,
  "dateOfBirth": "1990-01-01",
  "gender": 1
}
```

// Test Case 3: Invalid City

```
{
  "fullName": "Test User",
  "username": "test_invalid_city",
  "email": "invalidcity@test.com",
  "password": "Test@123",
  "cityId": 999, // ✗ City doesn't exist
  "dateOfBirth": "1990-01-01",
  "gender": 1
}
```

Parents/analytics

1 GET /api/Parents/analytics

Description

Retrieves **complete data** for all registered parents including user information, demographics, city details, and calculated metrics. Designed for data analytics, Power BI dashboards, and reporting purposes.

Endpoint Details

``http
GET /api/Parents/analytics

Authentication

None required (v1.0)

Request Headers

HTTP

Accept: application/json

Request Parameters

None

Response (200 OK - Success)

Response Structure

JSON

```
{
  "success": boolean,
  "message": string,
  "data": [
    {
      "userId": "guid",
      "fullName": "string",
      "username": "string",
      "email": "string",
      "userCreatedAt": "datetime",
      "userUpdatedAt": "datetime | null",
      "parentId": "guid",
      "dateOfBirth": "datetime",
      "age": integer,
      "gender": "string",
      "genderCode": integer,
      "cityId": integer,
      "cityName": "string",
      "cityNameAr": "string",
      "cityIsActive": boolean,
      "accountAgeInDays": integer,
      "ageGroup": "string",
      "region": "string"
    }
  ],
  "errors": []
}
```

Response Fields

Field	Type	Description	Example
success	boolean	Indicates if request was successful	true
message	string	Response message in Arabic	"تم استرجاع بيانات 5 ولي أمر بنجاح"
data	array	Array of parent analytics objects	[...]
errors	array	List of errors (empty on success)	[]

Parent Analytics Object Fields

Field	Type	Description	Example	Notes
userId	guid	User unique identifier	"3fa85f64-5717-4562-b3fc-2c963f66afa6"	Primary key from Users table
fullName	string	Parent's full name	"Ahmed Mohamed Ali"	Can be Arabic or English
username	string	Unique username	"ahmed2025"	Used for login
email	string	Email address	"ahmed@example.com"	Unique per user

userCreatedAt	datetime	Account creation date/time	"2025-11-10T08:30:15Z"	ISO 8601 format, UTC
userUpdatedAt	datetime/null	Last account update time	"2025-11-10T09:45:20Z" or null	Nullable, ISO 8601 format
parentId	guid	Parent unique identifier	"7fa85f64-5717-4562-b3fc-2c963f66afa7"	Primary key from Parents table
dateOfBirth	datetime	Parent's date of birth	"1985-03-20T00:00:00Z"	ISO 8601 format
age	integer	Current age in years	40	Calculated from dateOfBirth
gender	string	Gender text	"Male" or "Female"	Human-readable
genderCode	integer	Gender numeric code	1 or 2	1 = Male/Father, 2 = Female/Mother
cityId	integer	City identifier	1	Foreign key to Cities
cityName	string	City name (English)	"Cairo"	From Cities table
cityNameAr	string	City name (Arabic)	"القاهرة"	From Cities table

cityIsActive	boolean	City active status	true	Indicates if city is active
accountAgeInDays	integer	Days since registration	15	Calculated: now - createdAt
ageGroup	string	Age category	"36-45"	Calculated age bracket
region	string	Geographic region	"Greater Cairo"	Derived from city

Example Response

JSON

```
{
  "success": true,
  "message": "تم استرجاع بيانات 5 ولي أمر بنجاح",
  "data": [
    {
      "userId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "fullName": "Ahmed Mohamed Ali",
      "username": "ahmed2025",
      "email": "ahmed.mohamed@example.com",
      "userCreatedAt": "2025-10-26T12:30:45Z",
      "userUpdatedAt": "2025-11-05T08:15:30Z",
      "parentId": "7fa85f64-5717-4562-b3fc-2c963f66afa7",
      "dateOfBirth": "1985-03-20T00:00:00Z",
      "age": 40,
      "gender": "Male",
      "genderCode": 1,
      "cityId": 1,
      "cityName": "Cairo",
      "cityNameAr": "القاهرة",
      "cityIsActive": true,
      "accountAgeInDays": 15,
      "ageGroup": "36-45",
      "region": "Greater Cairo"
    }
  ],
}
```

```
{
  "userId": "4fa85f64-5717-4562-b3fc-2c963f66afa8",
  "fullName": "Fatima Hassan",
  "username": "fatima_hassan",
  "email": "fatima.h@example.com",
  "userCreatedAt": "2025-10-25T09:15:30Z",
  "userUpdatedAt": null,
  "parentId": "8fa85f64-5717-4562-b3fc-2c963f66afa9",
  "dateOfBirth": "1990-07-15T00:00:00Z",
  "age": 35,
  "gender": "Female",
  "genderCode": 2,
  "cityId": 2,
  "cityName": "Alexandria",
  "cityNameAr": "الإسكندرية",
  "cityIsActive": true,
  "accountAgeInDays": 16,
  "ageGroup": "26-35",
  "region": "Mediterranean"
},
"errors": []
}
```

Response (200 OK - No Data)

When no parents exist in the system:

JSON

```
{
  "success": true,
  "message": "لا توجد بيانات",
  "data": [],
  "errors": []
}
```

Response (500 Internal Server Error)

JSON

```
{
  "success": false,
  "message": "حدث خطأ في الخادم",
}
```

```
"data": null,
"errors": [
  "حدث خطأ غير متوقع أثناء استرجاع البيانات"
]
}
```

Age Groups Reference

Age Range	Age Group Code	Description
18-25	"18-25"	Young adults
26-35	"26-35"	Early career parents
36-45	"36-45"	Mid-career parents
46-55	"46-55"	Experienced parents
56+	"56+"	Senior parents

Regions Reference

Region	Cities Included
"Greater Cairo"	Cairo, Giza, Shubra El Kheima
"Mediterranean"	Alexandria

"Canal Zone"	Port Said
"Other"	Any other cities

Gender Codes

Code	Gender	Description
1	Male	Father
2	Female	Mother

Use Cases

1. **Power BI Dashboards** - Import all parent data for visualization
2. **Data Analytics** - Analyze demographics and trends
3. **Reports** - Generate parent statistics reports
4. **Data Export** - Export parent data to CSV/Excel
5. **Business Intelligence** - Track user growth and demographics

cURL Example

```
Bash

# Get all parents analytics data
curl -X GET
"[https://ajial-api-dev.azurewebsites.net/api/Parents/analytics](https://ajial-api-dev.azurewebsites.net/
api/Parents/analytics)" \
-H "Accept: application/json"
```

Python Example

```
Python

import requests

# Get all parents analytics data
```



```
response = requests.get(

'https://ajial-api-dev.azurewebsites.net/api/Parents/analytics')(https://ajial-api-dev.azurewebsites.net/
api/Parents/analytics)',
    headers={'Accept': 'application/json'}
)

if response.status_code == 200:
    data = response.json()
    print(f"Total parents: {len(data['data'])}")
    for parent in data['data']:
        print(f"{parent['fullName']} - {parent['cityName']} - Age: {parent['age']}")
else:
    print(f"Error: {response.status_code}")
```

Performance Notes

- **Response Time:** ~100-500ms (depending on data size)
- **Data Size:** ~1-2 KB per parent record
- **Recommended Max Records:** 10,000 parents (adjust if needed)
- **Caching:** Not implemented (v1.0)

Data Privacy & Security

Important Notes:

- Email addresses are included - ensure proper access controls in production
- Consider implementing authentication before production use
- Password hashes are NOT included in response
- Personal data - comply with GDPR/data protection regulations

GET /api/Parents/analytics/summary

Description

Retrieves **aggregated statistics and summary metrics** for all parents. Provides high-level insights including total counts, demographics distribution, and regional statistics. Ideal for dashboard KPIs and quick insights.

Endpoint Details

HTTP

GET /api/Parents/analytics/summary

Authentication

None required (v1.0)

Request Headers

HTTP

Accept: application/json

Request Parameters

None

Response (200 OK - Success)

Response Structure

JSON

```
{
  "success": boolean,
  "message": string,
  "data": {
    "totalParents": integer,
    "maleCount": integer,
    "femaleCount": integer,
    "averageAge": integer,
    "ageGroups": [
      {
        "ageGroup": string,
        "count": integer
      }
    ],
    "citiesDistribution": [
      {
        "city": string,
        "count": integer
      }
    ],
    "regionsDistribution": [
      {
        "region": string,
        "count": integer
      }
    ],
    "recentRegistrations": integer,
  }
}
```

```
"lastUpdated": datetime
}
}
```

Response Fields

Field	Type	Description	Example
success	boolean	Request success status	true
message	string	Response message (Arabic)	تم استرجاع الإحصائيات بنجاح
data.totalParents	integer	Total number of parents	150
data.maleCount	integer	Number of male parents (fathers)	85
data.femaleCount	integer	Number of female parents (mothers)	65
data.averageAge	integer	Average age of all parents	37
data.ageGroups	array	Distribution by age group	[...]
data.citiesDistribution	array	Distribution by city	[...]

data.regionsDistribution	array	Distribution by region	[...]
data.recentRegistrations	integer	Parents registered in last 30 days	12
data.lastUpdated	datetime	Timestamp of data retrieval	"2025-11-10T10:41:35Z"

Example Response

JSON

```
{
  "success": true,
  "message": "تم استرجاع الإحصائيات بنجاح",
  "data": {
    "totalParents": 150,
    "maleCount": 85,
    "femaleCount": 65,
    "averageAge": 37,
    "ageGroups": [
      {
        "ageGroup": "18-25",
        "count": 15
      },
      {
        "ageGroup": "26-35",
        "count": 52
      }
    ],
    "citiesDistribution": [
      {
        "city": "Cairo",
        "count": 65
      }
    ],
    "regionsDistribution": [
      {
        "region": "Greater Cairo",
        "count": 115
      }
    ]
  }
}
```

```
],  
  "recentRegistrations": 12,  
  "lastUpdated": "2025-11-10T10:41:35Z"  
}  
}
```
